Executive Summary

This assessment asks whether individual large-cap tech stocks offered a better opportunity than simply tracking the broader market. I compared NVIDIA and Intel with the Nasdaq Composite from January 2005 to December 2025, using one reproducible R script built on adjusted Yahoo Finance prices and Federal Reserve CPI data. I measured returns, volatility, downside risk, and long-run growth in both nominal and inflation-adjusted terms.

The outcomes could hardly be further apart. NVIDIA compounded at 39.2% a year and Intel at 5.0%, against 12.0% for the index. Inflation of 2.57% a year erodes all three further, but the gap barely closes. In 2005 purchasing power, $10,000 invested then would be worth $6.1 million in NVIDIA, $63,000 in the index, and just $16,000 in Intel. Both stocks were far riskier than the index with 46% and 31% annual volatility versus 18% – but only NVIDIA was actually compensated for it, with a Sharpe ratio of 0.90 against 0.63 for the index and a weak 0.26 for Intel. Risk isn't fixed: it shifts over time in phases of calm and turbulence and Intel's Value at Risk over the last two years is two-thirds worse than its twenty-one-year figure suggests.

For an investor, the index therefore belongs at the core of a portfolio and treat a single stock like NVIDIA as a small, deliberate side bet – not the main strategy. Decisions should be based on real, risk-adjusted returns rather than headline numbers, and risk should be re-checked using recent data rather than long-term averages.

Three limitations to my findings: NVIDIA was selected with hindsight, so the comparison shows what stock-picking costs when it fails more than it proves stock-picking works; historical VaR assumes the past repeats; and the Nasdaq excludes dividends while the stock series include them.